

The Price of Risk When Aggregation Fails: Relative Wealth Concerns with Heterogeneous Risk Aversion

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Abstract

Gómez (2007) shows that keeping-up-with-the-Joneses preferences with linear risk tolerance collapse into a rescaling of the representative agent's risk aversion, but only under restrictive conditions requiring common preference parameters. This paper characterises the failure, in a CARA economy where each investor compares his wealth to a weighted average over a network of peers. With common risk aversion the comparison is exactly neutral, for every influence network and every profile of comparison intensities, extending Gómez's CARA case to networks. With heterogeneous risk aversion the demand scales are a Katz–Bonacich centrality and the effective risk aversion depends on the network. In a mean-field limit it is $H(\theta + \lambda) - \lambda$, H the harmonic mean, so comparison interpolates aggregation from the harmonic to the arithmetic mean at a rate equal to the squared coefficient of variation of shifted risk tolerance. Away from that limit it need not be monotone: in a three-investor network the coefficient first rises, then falls below its no-comparison value. Reversal becomes rare as the population grows, but this reflects mixing rather than size. And homophily in risk aversion can suppress the effect entirely, the harmonic mean surviving at arbitrarily large comparison intensity.

Keywords: relative wealth concerns, keeping up with the Joneses, CARA aggregation, Katz–Bonacich centrality, homophily.

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1 Introduction

[10] studies an exchange economy in which agents evaluate consumption relative to a weighted average of the consumption of others, and asks when equilibrium prices depend

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only on aggregates. His Proposition 1 shows that linear risk tolerance with common cautiousness delivers a representative agent and rules out portfolio bias; his Proposition 2 gives the conditions under which prices depend on the aggregate endowment alone, and finds them restrictive: the preference parameter must be common across agents and, absent the trivial case of uniform weights, so must the equilibrium value of the endowment. Under those conditions the CARA case reduces cleanly, and keeping up with the Joneses is equivalent to lowering the representative agent’s risk aversion from α to $\alpha(1 - \gamma)$.

What happens outside those conditions is left open, and it is the natural question. [10] notes that in their absence equilibrium prices may be affected by the distribution of weights in the definition of the Joneses, by the preference parameters, and by the endowment distribution — three channels that a representative-agent reduction cannot express. This paper opens them, in the CARA case, for a benchmark that is agent-specific: investor a compares himself to $\sum_j A_{aj}W_j$, a weighted average over a network of influence rather than a single economy-wide aggregate.

The results divide according to whether risk aversion is common.

When it is, the comparison is exactly neutral. Every investor holds the position he would hold with no relative concern at all, for every row-stochastic A and every profile of comparison intensities λ_a (Theorem 1). This is the network extension of Gómez’s CARA case, and its content is that the network drops out entirely: no statistic of A survives into the equilibrium. It is worth stating separately because the proof is immediate and because it delimits precisely where the interesting region begins.

When risk aversion is heterogeneous, neutrality fails and the network matters. The demand scales solve $[\mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A]\vec{c} = \mathbf{1}$, a Katz–Bonacich system in the sense of [2], and we study it in three ways. Under a mean-field condition the effective market risk aversion is $H(\theta + \lambda) - \lambda$, so that comparison interpolates the aggregation of risk aversion from the harmonic mean at $\lambda = 0$ to the arithmetic mean as $\lambda \rightarrow \infty$, at a rate equal to the squared coefficient of variation of shifted risk tolerance (Proposition 1). Away from that condition the direction is not guaranteed and the relation need not be monotone: a three-investor network with a directed pattern of attention delivers a coefficient that first rises and then falls below its no-comparison value (Proposition 2). We measure how often that occurs under a dense i.i.d.-weight design, where it becomes rare as the population grows; this is a statement about mixing rather than about size, and block structure preserves reversal at any number of investors. And if investors compare themselves only to others of their own type, the aggregation does not move at all, however intense the comparison (Proposition 3); homophily in risk aversion blocks the mechanism completely.

Together these say that the comparison channel in this class of models is not a property of preferences but of the interaction between preference dispersion and network structure, and that the mean-field intuition — which is what a representative-agent treatment would deliver — can fail in both magnitude and sign.

2 Related literature

Preferences over relative consumption or wealth have been used to address a range of asset-pricing questions. [1] and [8] build consumption-based models with external benchmarks and obtain an amplified equity premium; [3] show how an external habit generates time-varying effective risk aversion; [4] make the wealth distribution a state variable under heterogeneous CRRA preferences, which is the dynamic counterpart of the configuration studied here. [7] provide the taxonomy under which (1) is jealousy combined with keeping up with the Joneses.

Within the CARA class, [5] study a two-agent economy with multiple risky assets and different risk tolerances, showing that relative concerns are equivalent to modified tolerances and lower equilibrium premia; that economy is Section 5 at $m = 2$. [6] obtain portfolio bias when the benchmark differs in composition from the investor's own holdings, a channel excluded here by the scalar structure of (2). [12] solve relative-performance games for CARA and CRRA investors in mean-field and n -agent form; their concern is the existence and structure of Nash equilibria rather than whether the aggregate moves, and their mean-field formulation is the analogue of our Proposition 1 in a different setting. [11] take the problem to incomplete markets in continuous time. [13] surveys what heterogeneity and inequality imply for asset prices more broadly.

Two connections deserve care. [9] gives the criterion governing the effect of endowment inequality on the equity premium, namely concavity of risk tolerance in wealth. Our (7) is an analogue for dispersion in a preference parameter rather than in endowments, and should be read as such rather than as an application of his theorem. And [2] give the linear-quadratic network game whose solution (3) reproduces; what the present setting adds is that the centrality vector becomes constant, and every statistic of the network drops out, precisely when risk aversion is common.

3 The economy

There are m investors, one riskless asset with net return r_f , and n risky assets with excess returns $\vec{y}$ that are Gaussian with mean $\vec{r}_e$ and non-singular covariance Σ . Investor a begins with wealth W_a^0 and chooses monetary positions $\vec{\pi}_a \in \mathbb{R}^n$, so that $W_a = W_a^0(1 + r_f) + \vec{\pi}_a^\top \vec{y}$. Preferences are

$$U_a = -\exp \left[-\theta_a W_a - \lambda_a \left(W_a - \sum_j A_{aj} W_j \right) \right], \quad \theta_a > 0, \lambda_a \geq 0. \quad (1)$$

Assumption 1 (Normalised benchmark). $A_{aa} = 0$, $A_{aj} \geq 0$ and $\sum_j A_{aj} = 1$ for every a .

The benchmark is thus a weighted average of the wealth of the other investors, as in [10], with the difference that the weights may vary across investors. Assumption 1 makes the preference invariant to a uniform translation of all wealths: adding a constant to every W_j shifts the exponent by $-\theta_a$ times that constant, independently of λ_a . It is

also what the results below require. Dropping it changes the object being compared: row sums below one describe comparison with a partial or external benchmark, which is a coherent alternative we do not study, while row sums above $1 + \theta_a/\lambda_a$ make an investor worse off when the whole economy becomes richer, which is harder to motivate. We retain Assumption 1 throughout and claim nothing about either case.

Assumption 2 (Nash behaviour). Each investor chooses $\vec{\pi}_a$ taking the positions $\vec{\pi}_j$, $j \neq a$, as fixed at their equilibrium values. An equilibrium is a profile $(\vec{\pi}_1, \dots, \vec{\pi}_m)$ and a vector $\vec{r}_e$ such that each $\vec{\pi}_a$ is optimal given the others and markets clear.

Assumption 2 is the standard notion and we state it explicitly because the conjecture it embodies matters for what follows. An investor behaving in this way conjectures that the positions of others do not respond to his own: the conjectural derivative is zero, and the conjectured levels are the equilibrium positions. This is distinct from an investor who believes that others hold whatever he holds, a possibility we take up separately in Section 6.

First-order conditions

Under (1) the random part of the exponent is $-(\theta_a + \lambda_a)\vec{\pi}_a - \lambda_a \sum_j A_{aj}\vec{\pi}_j^\top \vec{y}$. Writing $\vec{q}_a$ for the bracket, Gaussian returns give $\mathbb{E}[U_a] = -C \exp(-\vec{q}_a^\top \vec{r}_e + \frac{1}{2}\vec{q}_a^\top \Sigma \vec{q}_a)$, whose exponent is strictly convex in $\vec{\pi}_a$ because $\partial \vec{q}_a / \partial \vec{\pi}_a = (\theta_a + \lambda_a)\mathbb{I}$ is non-singular. The optimum is interior and solves $\vec{q}_a = \Sigma^{-1}\vec{r}_e$, that is

$$(\theta_a + \lambda_a)\vec{\pi}_a - \lambda_a \sum_j A_{aj}\vec{\pi}_j = \Sigma^{-1}\vec{r}_e. \quad (2)$$

Every solution is proportional to $\Sigma^{-1}\vec{r}_e$, so all investors hold the same portfolio of risky assets and differ only in scale; two-fund separation holds, as in Gómez's Proposition 1. Writing $\vec{\pi}_a = c_a \Sigma^{-1}\vec{r}_e$ reduces (2) to

$$[\mathbb{D}_{\theta+\lambda} - \mathbb{D}_\lambda A] \vec{c} = \mathbf{1}, \quad \mathbb{D}_x = \text{diag}(x_1, \dots, x_m), \quad (3)$$

whose solution is a Katz–Bonacich centrality of the influence network [2]. The matrix is strictly diagonally dominant for every positive θ_a , since the off-diagonal row sums are $\lambda_a \sum_j A_{aj} = \lambda_a < \theta_a + \lambda_a$; existence and uniqueness therefore hold, and $\vec{c} > 0$ follows from the Neumann expansion. Market clearing against a supply with value weights $\vec{w}$ and aggregate wealth W_M^0 gives

$$\vec{r}_e = \theta_M W_M^0 \Sigma \vec{w}, \quad \frac{1}{\theta_M} = \sum_a c_a, \quad \bar{\theta} = m \theta_M. \quad (4)$$

We call $\bar{\theta}$ the effective market risk-aversion coefficient and reserve *mean-covariance price-of-risk coefficient* for the product $W_M^0 \theta_M = \bar{W}^0 \bar{\theta}$, which is the dimensionless object; $\bar{\theta}$ alone carries units of inverse wealth and we do not call it a price.

Remark 1 (What the model determines). Equations (2)–(4) determine a mean–covariance relation given Σ and the supplies. They are not a complete system of asset prices: with payoffs and quantities as primitives a change in prices would move both the mean and the covariance of returns. Statements below concern demands and expected excess returns conditional on Σ , and nothing is claimed about price levels, market volatility or the invariance of betas.

4 Neutrality under common risk aversion

Theorem 1. *Let Assumptions 1 and 2 hold and suppose $\theta_a = \theta$ for every a . Then $\vec{c} = \theta^{-1}\mathbf{1}$ is the unique solution of (3), so every investor holds $\vec{\pi}_a = \Sigma^{-1}\vec{r}_e/\theta$ and $\bar{\theta} = \theta$, for every influence matrix satisfying Assumption 1 and every profile $(\lambda_1, \dots, \lambda_m)$.*

Proof. Substituting $\vec{c} = \theta^{-1}\mathbf{1}$ in row a of (3) and using $\sum_j A_{aj} = 1$ gives $(\theta + \lambda_a)/\theta - \lambda_a/\theta = 1$. Uniqueness follows from strict diagonal dominance, as above. $\square$

The cancellation occurs row by row, before any aggregation, which is why nothing about A or about the dispersion of λ_a enters. Economically, the benchmark is fixed in a unilateral deviation: under Assumption 2 the positions of others do not respond. Neutrality arises instead because at the equilibrium profile the benchmark carries the same risky exposure as the investor himself, so that $(\theta + \lambda_a)\vec{\pi} - \lambda_a A\vec{\pi} = (\theta + \lambda_a)\vec{\pi} - \lambda_a \vec{\pi} = \theta\vec{\pi}$. The benchmark exposure offsets the additional λ_a loading on own equilibrium wealth in the level of total risky exposure, not through any conjectured response to the deviation.

Theorem 1 is the network extension of the CARA case of [10], and we claim no more for it than that. His Proposition 2 identifies common cautiousness — here, common θ — as what makes the weighting drop out; the addition is that the weighting may be agent-specific and the comparison intensities may differ, and neither disturbs the conclusion.

Remark 2 (Relation to the keeping-up parametrization). The additive form (1) and the multiplicative form used by [10] describe the same preference. Collecting terms in the exponent,

$$-\theta_a W_a - \lambda_a \left(W_a - \sum_j A_{aj} W_j \right) = -\alpha_a \left(W_a - \gamma_a \sum_j A_{aj} W_j \right), \quad (5)$$

with $\alpha_a = \theta_a + \lambda_a$, $\gamma_a = \lambda_a/(\theta_a + \lambda_a)$ and hence $\theta_a = \alpha_a(1 - \gamma_a)$: α_a is the local coefficient of absolute risk aversion in own wealth and $\gamma_a \in [0, 1)$ the keeping-up weight. In this notation [10] writes the CARA case as $u(c, X) = -\exp(-\alpha(c - \gamma X))$ and reduces it to a representative agent with coefficient $\alpha(1 - \gamma)$, which is exactly θ ; his conclusion and Theorem 1 therefore agree wherever both apply.

The parametrizations are not interchangeable as *hypotheses*, however. His Proposition 2 requires the preference parameters to be identical across agents and, barring

uniform weights, the equilibrium value of the endowment to be identical as well. Theorem 1 requires neither. It constrains only the product $\alpha_a(1 - \gamma_a) = \theta$, so neutrality holds on the whole level set $\{(\alpha, \gamma) : \alpha(1 - \gamma) = \theta\}$, along which an investor with a stronger comparison motive is also locally more risk averse in own wealth; and initial wealth is absent from (2) altogether, since CARA demands have no wealth effects, so no restriction on the endowment distribution is needed. Reading θ_a rather than α_a as “the” risk aversion is what makes the statement look stronger than it is, and it is θ_a — the coefficient net of the comparison — whose dispersion drives everything in Section 5.

Remark 3 (Two notions of neutrality). Theorem 1 asserts that individual positions are unchanged, which is stronger than saying the effective coefficient is unchanged. The latter requires only $\sum_a c_a = m/\theta$, and deviations can offset. With four investors, $\theta = 1$ and demand scales $\bar{c} = (\frac{5}{7}, \frac{9}{7}, 1, 1)$ — corresponding to coefficients $\kappa = (\frac{7}{5}, \frac{7}{9}, 1, 1)$ — one has $\sum_a c_a = 4$ and hence $\bar{\theta} = \theta$ exactly, although the first two investors hold positions differing from the no-comparison benchmark by $-\frac{2}{7}$ and $+\frac{2}{7}$, or about 28.6 per cent in each direction. Results below concern $\bar{\theta}$, so the weaker notion is the relevant one, and statements about it should not be read as statements about allocations.

5 Heterogeneous risk aversion

We now let θ_a vary, which is exactly the configuration in which the aggregation conditions of [10] fail. The system (3) still has a unique positive solution, but it is no longer proportional to $\mathbf{1}$, and $\bar{\theta}$ depends on λ and on A .

5.1 The mean-field limit

Proposition 1. *Let $\lambda_a = \lambda$ for all a and consider a sequence of economies indexed by m in which the empirical distribution of θ_a converges weakly to a law with $\mathbb{E}[(\theta + \lambda)^{-1}] < \infty$ and $\{(\theta + \lambda)^{-1}\}$ uniformly integrable, and in which $\sum_j A_{aj}c_j - \bar{c}_m \rightarrow 0$ uniformly in a , where $\bar{c}_m = m^{-1} \sum_{a=1}^m c_a$. Then, writing $\bar{\theta}_m$ for the effective coefficient of the m -investor economy,*

$$\bar{\theta}_m(\lambda) \longrightarrow \bar{\theta}_\infty(\lambda) = H(\theta + \lambda) - \lambda, \quad H(x) = (\mathbb{E}[1/x])^{-1}. \quad (6)$$

The limit $\bar{\theta}_\infty$ is differentiable wherever the moments below exist, and

$$\frac{d\bar{\theta}_\infty}{d\lambda} = \frac{\text{Var}[(\theta + \lambda)^{-1}]}{\mathbb{E}[(\theta + \lambda)^{-1}]^2} \geq 0, \quad (7)$$

with equality if and only if θ is degenerate. If moreover $\mathbb{E}[\theta^3] < \infty$ then $\bar{\theta}_\infty = \mathbb{E}[\theta] - \text{Var}(\theta)/\lambda + O(\lambda^{-2})$ as $\lambda \rightarrow \infty$. Equation (7) is the derivative of the limit; we do not claim it is the limit of the derivatives of the finite economies.

Proof. Under the mixing condition row a of (3) becomes $(\theta_a + \lambda)c_a = 1 + \lambda\bar{c}$ in the limit, so $c_a = (1 + \lambda\bar{c})/(\theta_a + \lambda)$. Averaging and writing $M = \mathbb{E}[(\theta + \lambda)^{-1}]$ gives $\bar{c} = M/(1 - \lambda M)$

and $\bar{\theta} = 1/M - \lambda$. Differentiating gives (7), whose sign is Cauchy–Schwarz, with equality only for degenerate θ . The expansion follows from $\mathbb{E}[(\theta + \lambda)^{-1}] = \lambda^{-1}(1 - \mathbb{E}[\theta]/\lambda + \mathbb{E}[\theta^2]/\lambda^2 + O(\lambda^{-3}))$, which requires the third moment to control the remainder. $\square$

In this limit comparison interpolates the aggregation of risk aversion between the two classical aggregators: the harmonic mean at $\lambda = 0$, which is the standard CARA result, and the arithmetic mean as $\lambda \rightarrow \infty$. Since $H(\theta) \leq \mathbb{E}[\theta]$, the largest increase in the effective coefficient available is the harmonic–arithmetic gap of the risk-aversion distribution, a statistic of the cross-section rather than a free parameter.

Remark 4. The mixing condition is stated in terms of the endogenous vector $\vec{c}$, and is therefore close to assuming the conclusion. A primitive sufficient condition is that A have rows that are identical up to the leave-one-out correction, $A_{aj} = 1/(m-1)$ for $j \neq a$; then $\sum_j A_{aj}c_j = (\sum_k c_k - c_a)/(m-1)$ differs from $\bar{c}$ by $O(1/m)$ and (6) holds in the limit. Exact equality at finite m would require identical rows, which Assumption 1 forbids since $A_{aa} = 0$. What conditions on A short of this suffice is open, and in our view is the most useful direction in which this analysis can be extended.

5.2 Non-monotonicity and sign reversal

Proposition 2. *There exist normalised benchmarks and risk-aversion profiles for which $d\bar{\theta}/d\lambda < 0$ on a non-empty interval and for which $\bar{\theta}(\lambda) < \bar{\theta}(0)$ at sufficiently large comparison intensities.*

Proof. Take $m = 3$, $\theta = (\frac{1}{2}, 1, 2)$, $\lambda_a = \lambda$ and

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix},$$

which satisfies Assumption 1. Solving (3) gives

$$\bar{\theta}(\lambda) = \frac{3(\lambda + 2)(3\lambda + 1)}{12\lambda^2 + 24\lambda + 7}, \quad \bar{\theta}'(\lambda) = -\frac{3(12\lambda^2 + 6\lambda - 1)}{(12\lambda^2 + 24\lambda + 7)^2}. \quad (8)$$

Hence $\bar{\theta}'(0) = 3/49 > 0$ and $\bar{\theta}'(\lambda) < 0$ for $\lambda > (\sqrt{21} - 3)/12 \approx 0.132$. The function rises from $\bar{\theta}(0) = 6/7 \approx 0.857$ to a maximum ≈ 0.860 at that point, then falls, returning to $6/7$ at $\lambda = 1/3$ and continuing downwards: $\bar{\theta}(1) = 0.837209$ and $\bar{\theta}(2) = 0.815534$. $\square$

The example is more informative than a monotone decrease would be. Comparison first raises the effective coefficient, as the mean-field intuition of Proposition 1 suggests, and only at higher intensity does the structure of attention take over and reverse the direction. The pattern is what produces it: the most risk averse investor looks at the least risk averse, while the other two look only at each other.

Reversal is not generic. Under the dense i.i.d.-weight design described below it becomes rare as the number of investors grows and local benchmark exposures approach

complete mixing; Remark 5 shows that this is a property of that design and not of size. Table 1 reports two distinct frequencies over 20,000 independent draws at each size: the *cumulative* event $\bar{\theta}(0.5) < \bar{\theta}(0)$, and the *marginal* event $\bar{\theta}'(0.5) < 0$. The derivative is computed analytically rather than by finite differences: writing $M(\lambda) = \mathbb{D}_\theta + \lambda(\mathbb{I} - A)$ and $\vec{c} = M^{-1}\mathbf{1}$,

$$\vec{c}' = -M^{-1}(\mathbb{I} - A)\vec{c}, \quad \bar{\theta}' = -\frac{m \mathbf{1}^\top \vec{c}'}{(\mathbf{1}^\top \vec{c})^2}. \quad (9)$$

The number of investors is held fixed within each row, so that the comparison across rows identifies the effect of size. Off-diagonal entries of A are uniform on $(0, 1)$ before row normalisation, and $\theta_a = \max\{1 + \sigma Z_a, 0.15\}$ with Z_a standard normal.

m	σ	cumulative: $\bar{\theta}(0.5) < \bar{\theta}(0)$		marginal: $\bar{\theta}'(0.5) < 0$	
		frequency	95% interval	frequency	95% interval
3	0.4	0.0919	[0.0879, 0.0959]	0.0952	[0.0912, 0.0993]
3	0.8	0.0340	[0.0316, 0.0366]	0.0372	[0.0346, 0.0399]
6	0.4	0.0057	[0.0047, 0.0068]	0.0062	[0.0052, 0.0073]
6	0.8	0.0008	[0.0005, 0.0014]	0.0008	[0.0005, 0.0012]
10	0.4	0.0001	[0.0000, 0.0004]	0.0001	[0.0000, 0.0004]
10	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
20	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.4	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]
60	0.8	0.0000	[0.0000, 0.0002]	0.0000	[0.0000, 0.0002]

Table 1: Frequency of negative cumulative and marginal effects at $\lambda = 0.5$, over 20,000 draws per row with Wilson intervals. All ten cells of the design are reported; entries of 0.0000 are exact zeros, no draw out of 20,000 producing the event, and 0.0002 is the upper endpoint of the 95% Wilson interval when no event is observed. Under this design reversal is already below one per cent at $m = 6$ and indistinguishable from zero at $m \geq 10$. It is also rarer at higher dispersion of risk aversion, which is consistent with the positive mean-field effect of (7) growing with dispersion and so becoming harder for network asymmetry to overturn. The decay in m is specific to the dense i.i.d.-weight ensemble; see Remark 5.

Remark 5 (Reversal at any population size). The decay in Table 1 reflects mixing, not size. Because (3) is local, a block-diagonal A decouples: each block solves its own system and $\bar{\theta} = (m^{-1} \sum_a c_a)^{-1}$ is the reciprocal of the population-average capacity. Replicating an identical block leaves that average, and hence $\bar{\theta}$, unchanged. Taking k disjoint copies of the network of Proposition 2 therefore gives $\bar{\theta}(\lambda)$ independent of k , so that $\bar{\theta}(1) = 0.837209 < 6/7 = \bar{\theta}(0)$ at $m = 3k$ for every k ; we verify this numerically up to $m = 3000$. The construction is robust to weak ties across blocks: mixing each row with a fraction ϵ of the uniform complete network preserves the reversal at $m = 30$ for $\epsilon \leq 0.1$ ($\bar{\theta}(1) = 0.8526$ at $\epsilon = 0.1$) and destroys it by $\epsilon = 0.2$ ($\bar{\theta}(1) = 0.8681$). Table 1

should therefore be read as describing a dense ensemble in which local benchmarks approach a common average, and not as a claim about large networks in general: block structure, or any persistent correlation between type and network position, sustains the reversal at arbitrary population size.

Note the contrast with Proposition 3, which uses the same block structure to the opposite end. There risk aversion is constant *within* blocks and the mechanism vanishes; here it is heterogeneous within blocks and absent *across* them, and the mechanism survives at any scale. What separates the two is where the heterogeneity sits relative to the partition, not the topology.

5.3 Homophily blocks the mechanism

Proposition 3. *Partition the investors into groups such that $A_{aj} = 0$ whenever a and j belong to different groups, and suppose θ_a is constant within each group. Then $c_a = 1/\theta_a$ solves (3) for every profile $(\lambda_1, \dots, \lambda_m)$, and $\bar{\theta}$ equals the harmonic mean of θ regardless of the intensity of comparison.*

Proof. Row a of (3) reads $(\theta_a + \lambda_a)c_a - \lambda_a \sum_j A_{aj}c_j = 1$. If $c_j = 1/\theta_j$ and $\theta_j = \theta_a$ for every j with $A_{aj} > 0$, the left-hand side is $(\theta_a + \lambda_a)/\theta_a - \lambda_a/\theta_a = 1$. Uniqueness gives the claim, and $1/\theta_M = \sum_a 1/\theta_a$ is the harmonic mean up to the factor m . $\square$

With two equally sized groups at $\theta \in \{0.5, 2\}$, $\bar{\theta} = 0.8$ at $\lambda = 0$, at $\lambda = 1$ and at $\lambda = 1000$; the arithmetic mean 1.25 that Proposition 1 would predict is never approached. Assortative matching on risk aversion therefore removes the comparison channel entirely, which is worth knowing because assortativity on wealth and on financial sophistication is a documented feature of social networks.

6 An aside on projection beliefs

Assumption 2 has investors conjecture that others do not respond to their own trade. A different behavioural assumption, sometimes invoked under the heading of the false consensus effect [14], is that an investor believes others hold what he holds. This is not a variant of Nash behaviour — the conjectured positions are not the equilibrium ones, and are in general not realised — and we treat it separately.

Suppose investor a believes that every j in his benchmark holds the same portfolio *weights* as he does, so that $\hat{\pi}_{j|a} = (W_j^0/W_a^0)\bar{\pi}_a$. Substituting in (1) and optimising gives

$$\kappa_a \bar{\pi}_a = \Sigma^{-1} \bar{r}_e, \quad \kappa_a = \theta_a + \lambda_a \left(1 - \frac{\bar{W}_a^0}{W_a^0} \right), \quad \bar{W}_a^0 = \sum_j A_{aj} W_j^0, \quad (10)$$

provided $\kappa_a \neq 0$. The coefficient now depends on the investor's position in the wealth distribution: an investor poorer than his benchmark has $\kappa_a < \theta_a$ and behaves as though less risk averse.

Two observations. First, the units of the projection decide the outcome: an investor who projects his *monetary* position rather than his weights believes $\hat{\pi}_{j|a} = \vec{\pi}_a$, which returns $\kappa_a = \theta_a$ and neutrality. The psychological description “he assumes others are like him” does not by itself determine the equilibrium. Second, κ_a may vanish or turn negative for investors sufficiently poor relative to their benchmark. The *perceived* individual optimisation problem remains well posed for $\kappa_a \neq 0$ — the exponent is strictly convex and the optimum interior, though optimality is relative to a benchmark that is not the one his peers actually realise — but the position is then a short one, and the aggregation (4) mixes terms of both signs, so that $\sum_a \kappa_a^{-1}$ can vanish or change sign and no vector of expected returns clears the market with a positive premium. This is a restriction on admissible configurations rather than a failure of individual optimisation.

We do not develop this case further. It rests on beliefs that are not realised in equilibrium, and a satisfactory treatment would require a decision-theoretic statement of what the investor maximises when his perceived benchmark differs from the positions his peers actually choose, together with a welfare criterion that does not presume the beliefs are correct.

7 Conclusion

Where the aggregation conditions of [10] hold, relative wealth concerns in a CARA economy are neutral: with common risk aversion every investor holds the position he would hold without them, whatever the network of influence and whatever the dispersion of comparison intensities. Where the conditions fail, because risk aversion is heterogeneous, the network enters and the effect is governed by the interaction between preference dispersion and network structure. Under complete mixing the comparison interpolates the aggregation of risk aversion from the harmonic mean to the arithmetic mean, at a rate given by the squared coefficient of variation of shifted risk tolerance. Away from it, the direction is not guaranteed: a small directed network can make the relation non-monotone and eventually drive the coefficient below its no-comparison value. Under a dense i.i.d.-weight design this becomes rare as the population grows, but that reflects the approach to complete mixing rather than size as such: with block structure the same reversal survives at any number of investors. Assortative matching on risk aversion can suppress the effect altogether.

The practical reading is that a representative-agent treatment of relative concerns is safe exactly where Gómez’s conditions hold and unreliable elsewhere, in a way that depends on features of the interaction structure that a representative agent cannot express. Characterising the classes of networks for which the mean-field direction is guaranteed — reversibility, double stochasticity, bounds on assortativity between risk aversion and centrality — would make the result usable, and is the obvious next step.

Replication

All numerical results are reproducible from the repository

<https://github.com/renatovicente/price-of-risk-aggregation>

The module `scripts/network.py` solves (3); `scripts/neutrality.py` verifies Theorem 1 and Proposition 1; `scripts/sign_reversal.py` produces Table 1, including the network and type generators and the seeds; `scripts/block_reversal.py` produces the block replication and the mixing robustness of Remark 5. The scripts depend only on `numpy` and `scipy`, read no data and draw from fixed seeds; `./reproduce.sh` runs all four in about fifteen seconds.

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